MIL-C-50471 (MU)
14 January 1970
Superseding
(See para. 6.3.6)

MILITARY SPECIFICATION

CHARGE, SUPPLEMENTARY ASSEMBLY WITH SPACER FOR DEEP CAVITY LOADED PROJECTILE, LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for supplementary charge assembly for deep cavity loaded projectiles.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168- Lot Numbering of Ammunition
MIL-STD-1169- Packing and Marking for Shipment of Inert Ammunition Components: General Specification for
MIL-STD-1235- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

FSC: 1315

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DRAWINGS

ARMY

8797088	-	Spacer
8797090	-	Supplementary Charge Assembly
8861645	-	Carton, Packing, Ammunition, for Supplementary Charge for Deep Cavity Loaded Shell Assembly, Details and Packing
8861646	-	Box, Packing, Ammunition, for Supplementary Charge for Deep Cavity Projectiles

PUBLICATIONS

ARMY

ET-8797090- Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Parts and assemblies.-The parts and assemblies shall comply with all requirements specified on drawing (dwg) 8797088 and 8797090 as applicable and with all requirements specified in applicable specifications.

3.3 Workmanship.-All parts shall be dry and free of chips, burrs, dirt, grease, rust and other foreign material. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein,

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unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.

b. Number of units of product inspected.

c. Results obtained, by defect code, for all inspections performed.

d. Drawings, specification number and date, together with an identification and date of changes.

e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions

4.2.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot which is defined as an essentially homogeneous collection of units or product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer in one unchanged process, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions.

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Changes to the process, specification, or drawing not affecting safety, performance, interchangeability or storage as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105. Unless otherwise approved by the contracting officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots of components or subassemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain:

a. TNT of one type from one lot interfix number from one manufacturer.

b. Bodies of one lot interfix number from one manufacturer.

4.2.2 Examination.-Inspection for Critical defects (and Major defects, when so specified), shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspection listed shall be in accordance with 4.2.4.

4.2.2.1 Body, prior to loading (see dwg. 8838201 a detail of dwg. 8797090).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Crack, split, puncture or similar damage extending through material.....	Visual	01001
Minor:	AQL 2.5 percent		
201.	Damage (not extending through material) or distortion.....	Visual	01006
202.	Marking missing, misleading or unidentifiable.....	Visual	01007

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203.	Evidence of poor workmanship (see 3.3)	Visual	01008

4.2.3.2 Disc closing, prior to loading (see dwg. 8838202 a detail of dwg. 8797090)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Crack, split, puncture or similar damage extending through material	Visual	02002
Minor:	AQL 1.0 percent		
201.	Damage (not extending through material) or distortion	Visual	02004
202.	Evidence of poor workmanship (see 3.3)	Visual	02005

4.2.3.3 Spacer (see dwg. 8797088)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 1.5 percent		
201.	Outside diameter	Gage	04001
202.	Inside diameter	Gage	04002
203.	Total length	Gage	04003
204.	Evidence of poor workmanship (see 3.3)	Visual	04004

4.2.3.4 Supplementary charge assembly (without pad and tab) (see dwg. 8797090)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Total length	Gage	05001

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Minor: None defined

4.2.2.5 Supplementary charge assembly (see dwg. 8797090)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Pad missing	Visual	06001
Major:	100 percent inspection		
102.	Pellets loose	Manual	06002
Major:	AQL 0.40 percent		
101.	Outside diameter, max,	Gage	06003
Minor:	AQL 2.5 percent		
201.	Tab missing, incorrect or inadequate	Visual	06004
202.	Crimp not full 360 degrees	Visual	06005
203.	Wrinkled body	Visual	06006
204.	Evidence of poor workmanship (see 3.3)	Visual	06007
205.	Marking missing, misleading or unidentifiable	Visual	06008
206.	Pad improperly assembled or aligned	Visual	06009
207.	Explosive on outside surface	Visual	06010

4.2.2.6 Carton, prior to closing (see dwg. 8861645)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 1.0 percent		
201.	Separator damaged to the extent that the supplementary charges touch each other	Visual	07001
202.	Supplementary charge missing	Visual	07002

4.2.2.7 Carton, sealed (dwg. 8861645)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			

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Major:	AQL 0.40 percent		
101.	Carton damaged to the extent that the contents are exposed or likely to become exposed in subsequent handling	Visual	08001
102.	Sealing strip torn, incomplete or not securely sealed to the carton	Visual	08002
Minor: None defined			

4.2.2.8 Wood packing box (prior to closing) (see dwg. 8861646).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Case liner incorrect, incomplete or insecurely sealed	Visual	09001
Minor: None defined			

4.2.2.9 Wood packing box sealed (see dwg. 8861646).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Box damaged to the extent that the contents are exposed or likely to become exposed in subsequent handling	Visual	10001
102.	Hardware or strapping broken or loose	Visual	10002
Minor:	AQL 1.5 percent		
201.	Hardware or strapping incorrectly assembled	Visual	10003
202.	Car seal unsealed or improperly positioned	Visual	10004
203.	Contents loose	Visual	10005
204.	Marking missing, misleading or unidentifiable	Visual	10006

4.2.3 Testing

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4.2.3.1 Specific gravity of charge (see dwg. 8797090), Major defect.

4.2.3.1.1 Pellets.-Ten randomly selected TNT pellets shall be tested from each eight hours of production. Failure to comply with the applicable requirements shall be cause for rejection of the eight hours production represented by the sample. Testing shall be performed as specified in 4.3.1.1, Code No. 11001.

4.2.3.1.2 Cast.-One body from each one hundred to be loaded shall be selected and marked sample. Failure to comply with the applicable requirements shall be cause for rejection of the lot. Testing shall be performed as specified in 4.3.1.2, Code No. 11002.

4.2.3.2 Barium stearate content of charge (see dwg. 8797090)-Major defect (applicable to pellets).-Five grams of material shall be randomly selected just prior to pelleting, from each eight hours of production or portion thereof. Failure to comply with the applicable requirement shall be cause for rejection. Testing shall be performed as specified in 4.3.2, Code No. 12001.

4.2.4 Inspection Equipment.-Equipment Tabulation Number ET-8797090 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 Government right to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the ET shall be in accordance with Specification MIL-I-45607.

4.2.4.3 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures.

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4.3.1 Specific gravity.

4.3.1.1 Pellets.-A beaker shall be filled with clean mercury. A semi-hard rubber perforated plate, pierced at each corner with sharp pins so that one half inch of the pin protrudes from each side, shall be placed in the mercury and the mercury shall be leveled off by pressing a glass plate until it is even with the top of the beaker. The pellet shall be placed beneath the prongs of the rubber plate and the mercury shall be leveled as before. The excess mercury shall be caught in a clean evaporating dish, and the mercury in the dish weighed. The specific gravity shall be calculated as follows:

Specific gravity = 13.54 w/w1

where:

w = weight of sample
w1 = weight of displaced mercury

4.3.1.2 Cast.-The sample body shall be weighed and then filled with water and weighed. The sample body shall be dried and then placed in the loading line and loaded under conditions identical in all respects with those employed in loading the lot. After loading, each sample body shall be weighed, then filled with water and reweighed. The specific gravity shall be calculated as follows:

Specific gravity = (C - A) / ((B - A) - (D - C))

where:

A = weight of empty body
B = weight of body filled with water
C = weight of body and TNT
D = weight of body and TNT and water

4.3.2 Barium stearate.-An accurately weighed portion of 5 grams shall be transferred to a tared, medium porosity, sintered glass filtering crucible. The sample shall be extracted with four 10-milliliter portions of acetone. Each portion shall remain in contact with the residue for approximately five minutes before applying suction. Any lumps which may persist shall be broken up and the sides of the crucible shall be washed down with each 10 milliliter portion of acetone to insure complete removal of the TNT. The residue shall be aspirated until the odor of acetone is no longer detectable. Collect the fourth extract in a clean container, and add a few drops of approximately N/10 sodium hydroxide solution. The absence of a red color indicates complete extraction. If a red color is produced, extract with additional 10 ml portions of acetone until complete extraction is indicated. The crucible and its contents shall be placed in an

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oven maintained at 105 degrees Centigrade (°C) plus or minus 5°C for one-half hour, cooled to room temperature in a desiccator and then weighed. The percent barium stearate shall be calculated as follows:

Percent barium stearate = 100 A/B

where:

A = weight of residue in crucible
B = weight of specimen

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and marking.

5.1.1 Level C.-The supplementary charges (loaded) shall be preserved and packaged in accordance with dwg. 8861645.

5.2 Packing and Marking

5.2.1 Level C.-The cartons (loaded) shall be packed in accordance with dwg. 8861646.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.-The digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the Inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists (EL).-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA-1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated

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in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment lists codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD-Contractor design on controlled contractor format and/or commercial equipment.
MU-Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approval practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflected the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility

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and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the Contracting Officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.3.6 Supersession data.-This specification includes the requirements of MIL-C-46684(MU) dated 11 October 1968 for Loading, Assembling, and Packing for Supplementary Charge for Deep cavity loaded Projectile.

Custodian:
ARMY-MU

Preparing Activity:
ARMY-MU

Project Number: 1315-A-686

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-2379

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